Predicting Delivery Delays Using Machine Learning in Telecom EPC Operations

An MBA Case Study focused on Nile For Integrated Solutions (NIS) in Egypt.

Candidate ID: 23227118

THE PROBLEM

A 20.5% Service Level Agreement (SLA) breach rate in field communications logistics causes heavy financial and operational leakage.

THE SOLUTION

An event-driven architecture combining an XGBoost predictive model, Odoo ERP integration, and Robotic Process Automation (RPA).

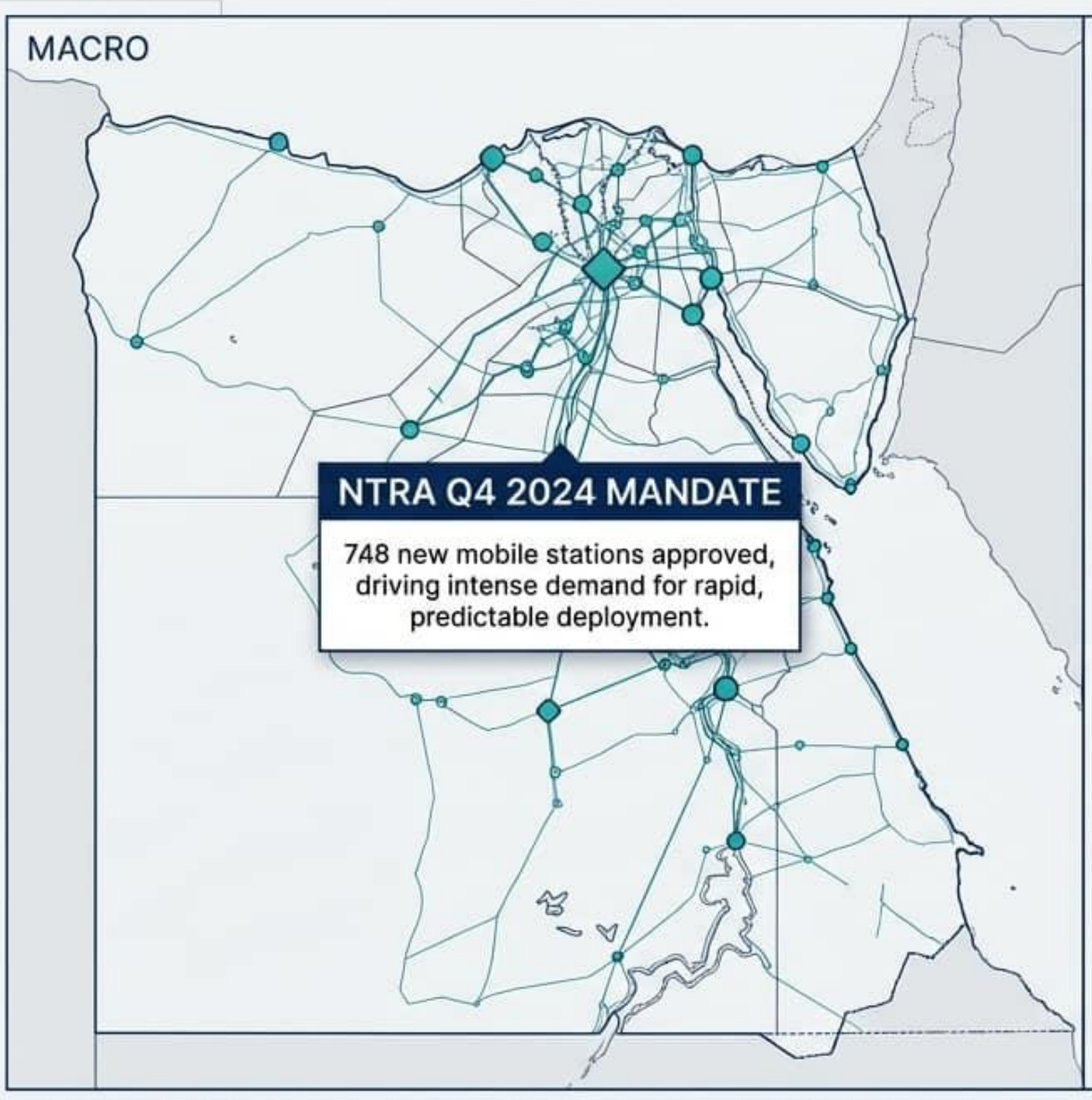
THE IMPACT

Projected reduction to a 12.3% breach rate (1,199 avoided breaches) yielding a 157% Year-1 ROI.

NotebookLM

EXECUTION SPEED IS A STRATEGIC MANDATE IN EGYPT'S EXPANDING TELECOM SECTOR

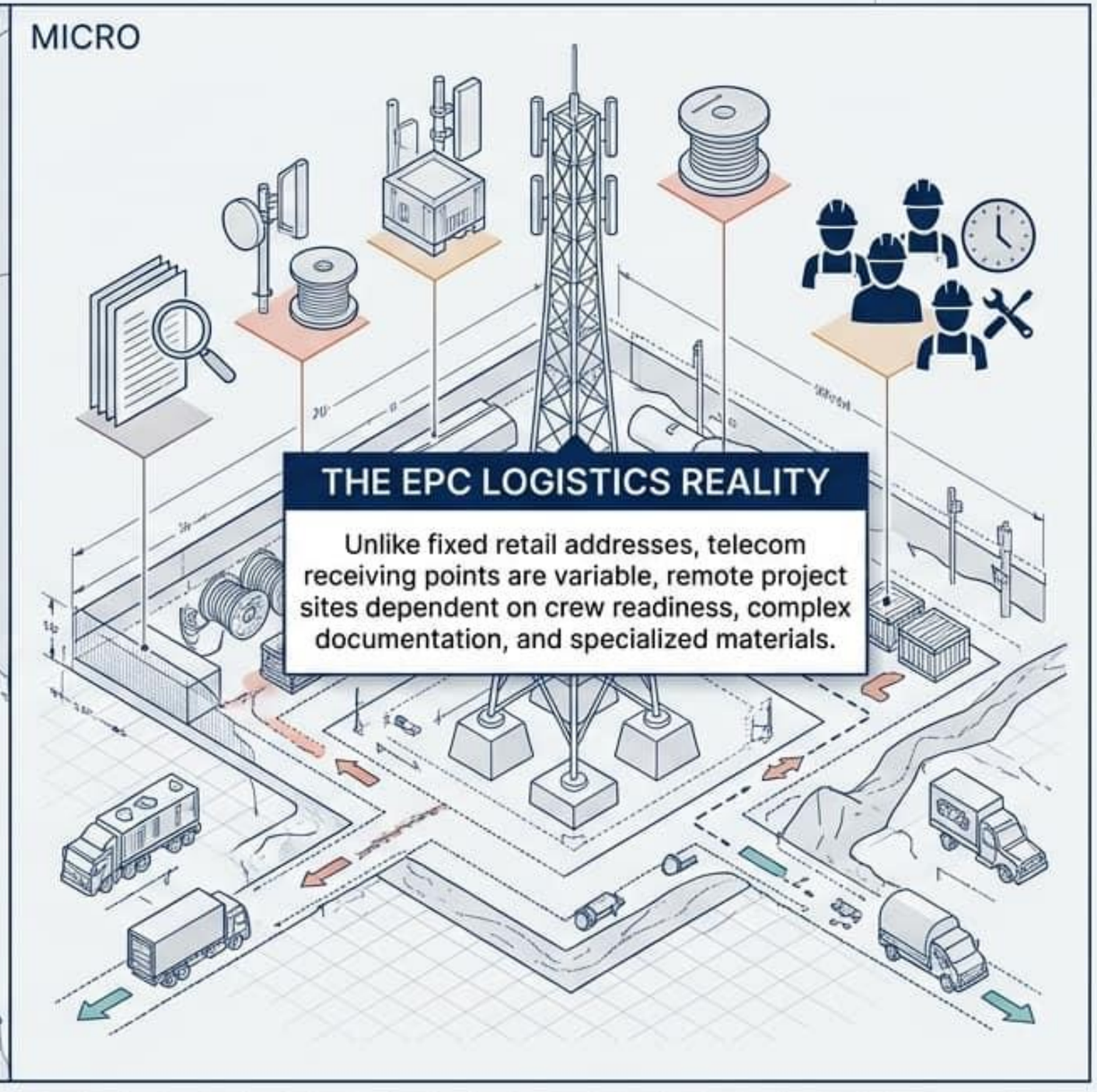
MACRO



NTRA Q4 2024 MANDATE

748 new mobile stations approved, driving intense demand for rapid, predictable deployment.

MICRO

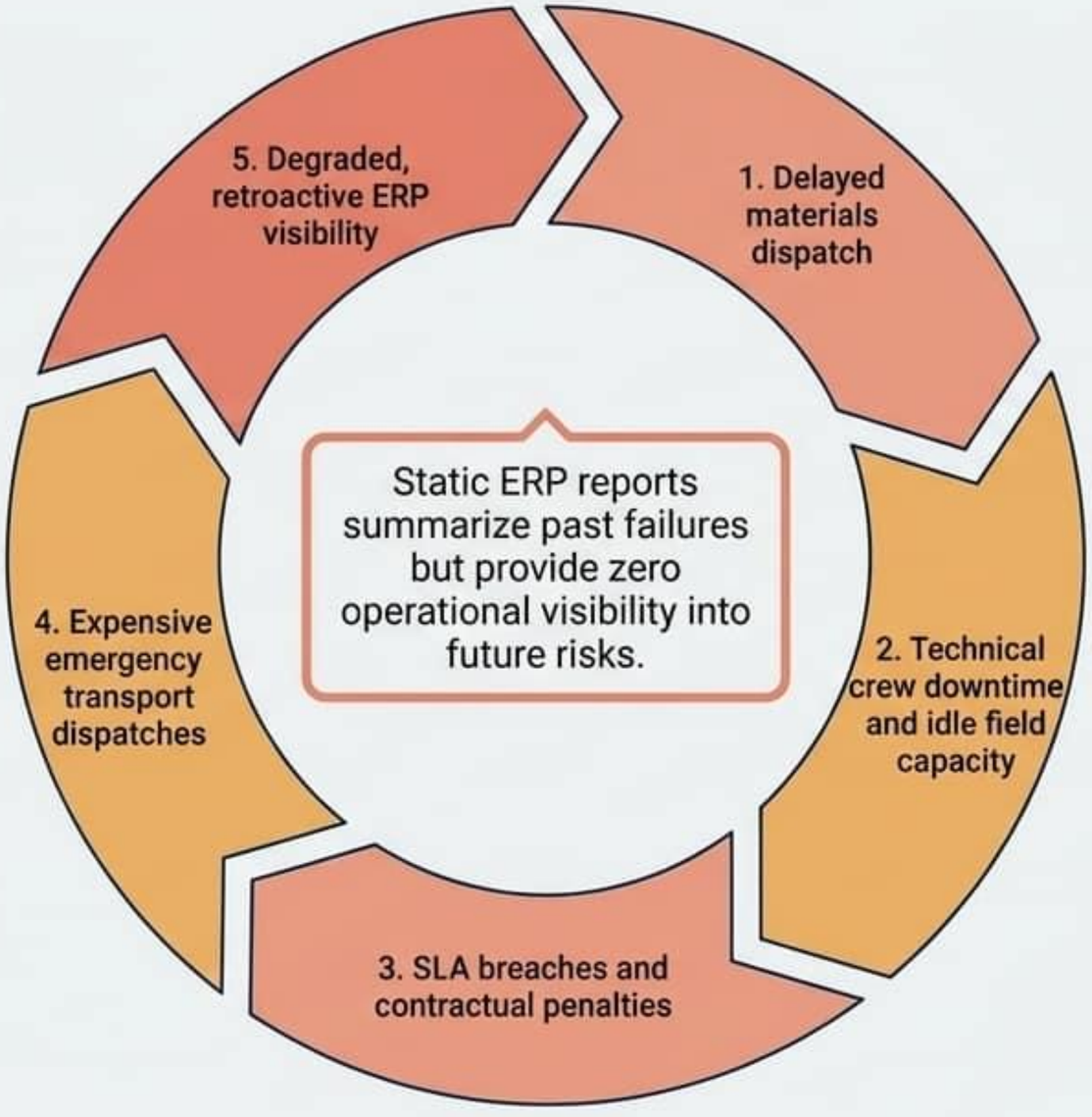


THE EPC LOGISTICS REALITY

Unlike fixed retail addresses, telecom receiving points are variable, remote project sites dependent on crew readiness, complex documentation, and specialized materials.

NotebookLM

TRADITIONAL DECISION-MAKING CREATES A COSTLY CYCLE OF LOGISTICAL BOTTLENECKS



THE BASELINE REALITY

~40
Daily Orders

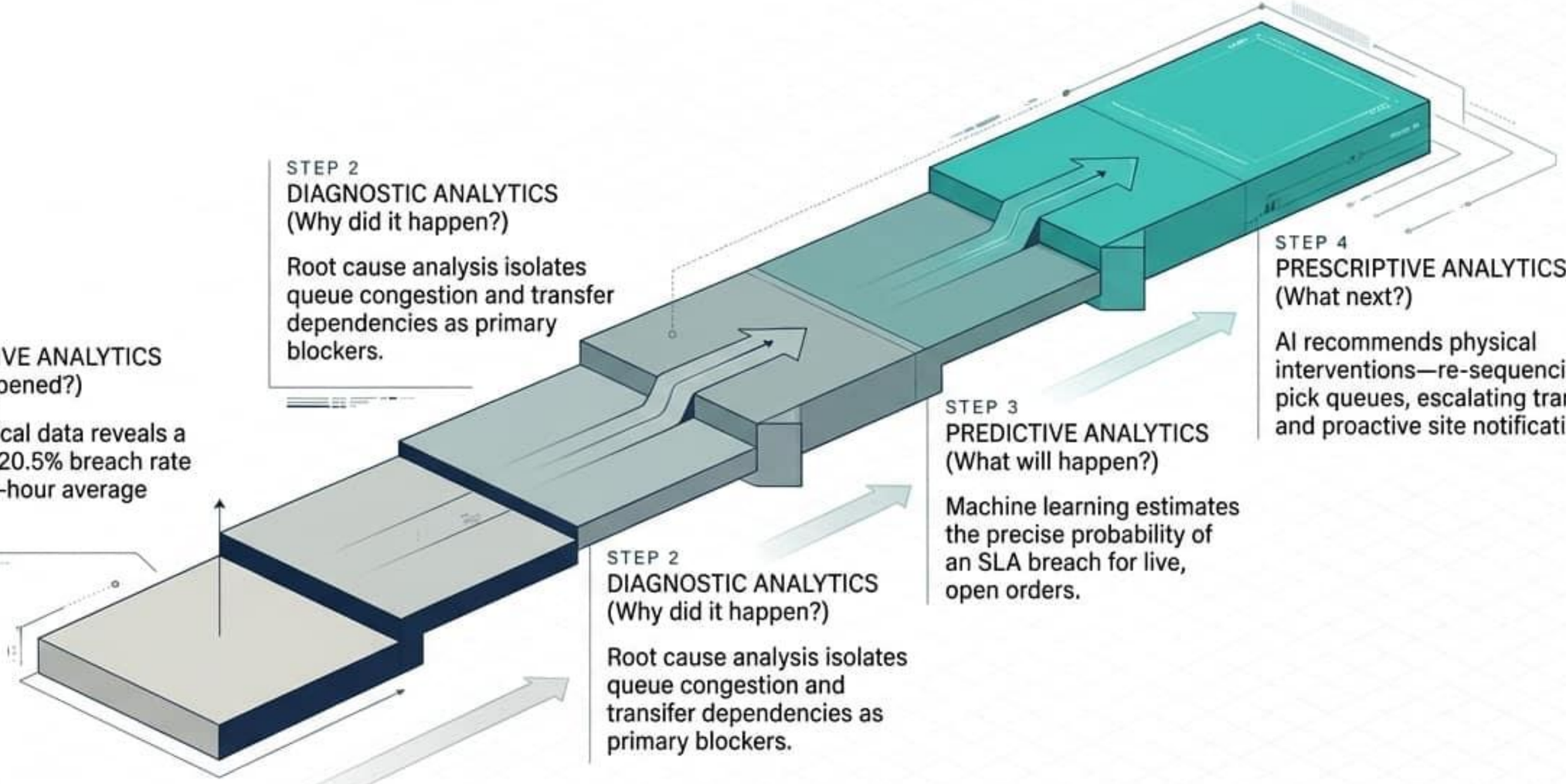
14,600
Annual Delivery Records

20.5%
Current Overall SLA Breach Rate

3.68 hrs
Average Lead Time

NotebookLM

VALUE CREATION REQUIRES MOVING FROM PASSIVE REPORTING TO ACTIVE DECISION DESIGN



STEP 1
DESCRIPTIVE ANALYTICS
(What happened?)

ERP historical data reveals a persistent 20.5% breach rate and a 3.68-hour average lead time.

STEP 2
DIAGNOSTIC ANALYTICS
(Why did it happen?)

Root cause analysis isolates queue congestion and transfer dependencies as primary blockers.

STEP 2
DIAGNOSTIC ANALYTICS
(Why did it happen?)

Root cause analysis isolates queue congestion and transifer dependencies as primary blockers.

STEP 3
PREDICTIVE ANALYTICS
(What will happen?)

Machine learning estimates the precise probability of an SLA breach for live, open orders.

STEP 4
PRESCRIPTIVE ANALYTICS
(What next?)

AI recommends physical interventions—re-sequencing pick queues, escalating transfers, and proactive site notification.

NotebookLM

STATISTICAL ACCURACY RELIES ON CAPTURING TRUE OPERATIONAL CONSTRAINTS

ANATOMY OF AN ORDER

SLA TIGHTNESS (Numeric)

The precise delta between promised operational time and baseline transit expectation.

DISTANCE BUCKET (Categorical)

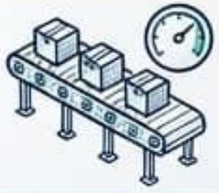
Captures true route complexity rather than raw kilometers (crucial for routes >220km).

ERP Order Document

INTER-WAREHOUSE TRANSFER (Binary)

Flags crippling dependencies on external stock movement.

PICK QUEUE (Numeric)



Measures real-time warehouse workload at the exact moment of order release.

SYNTHESIS: Distance alone is a weak predictor; SLA tightness combined with real-time queue congestion provides the definitive risk signal.

NotebookLM

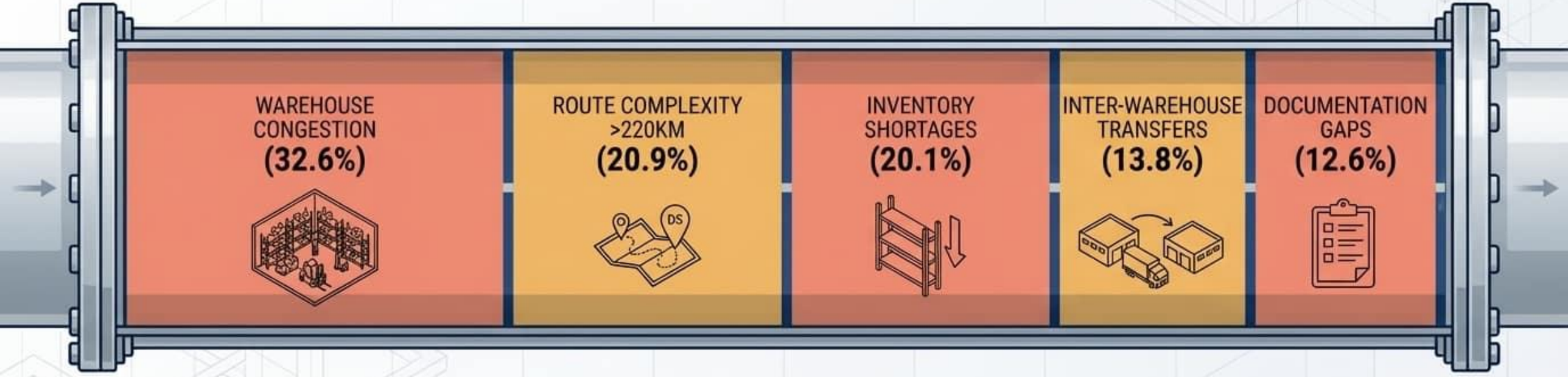
XGBoost provides the optimal balance of risk detection and operational credibility

MODEL	STATISTICAL METRICS	BUSINESS UTILITY
Logistic Regression	Recall: 0.864 / Precision: 0.579	Interpretable baseline, but causes severe alert fatigue due to excessive false positives.
Random Forest	ROC-AUC: 0.914 / Accuracy: 0.850	Non-linear benchmark. Respectable, but trails gradient boosting in discrimination.
XGBoost (Selected)	ROC-AUC: 0.936 / Accuracy: 0.889 / Precision: 0.757	Highly credible early-warning system with manageable false alarms.

Explainability Mandate: XGBoost ranks specific drivers (Distance, Transfers, Backorders) to ensure management accountability, preventing a 'black box' score.

NotebookLM

Delivery delays represent cross-functional failures rather than simple transport issues

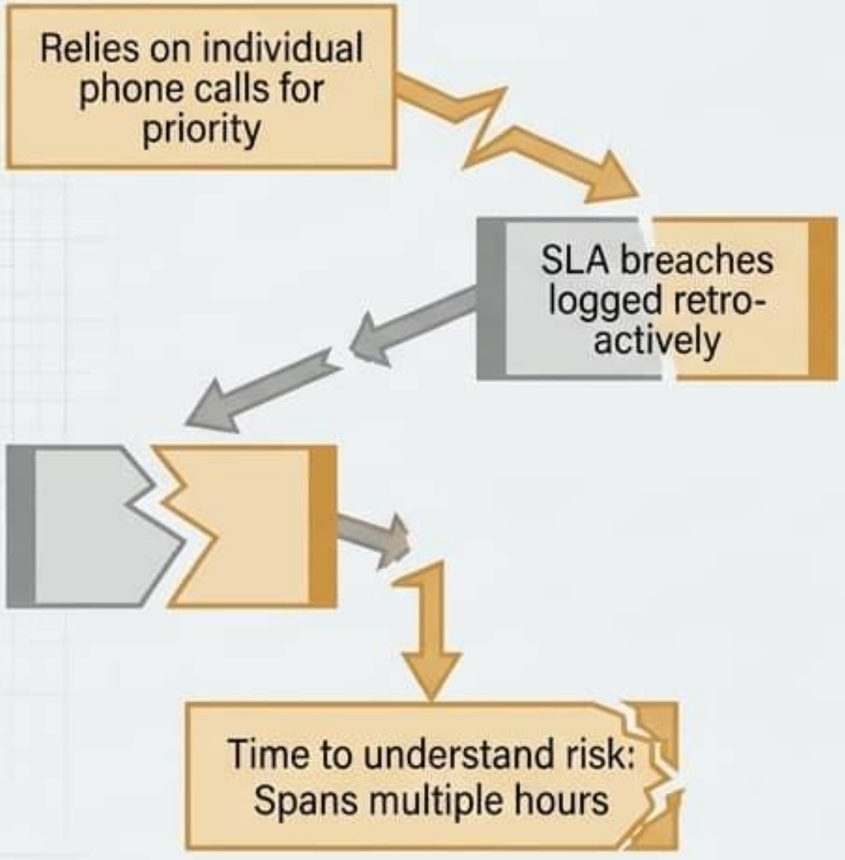


STRATEGIC INSIGHT:
Solving the SLA crisis requires earlier cross-functional coordination between stock allocation and project scheduling, not merely acquiring faster trucks.

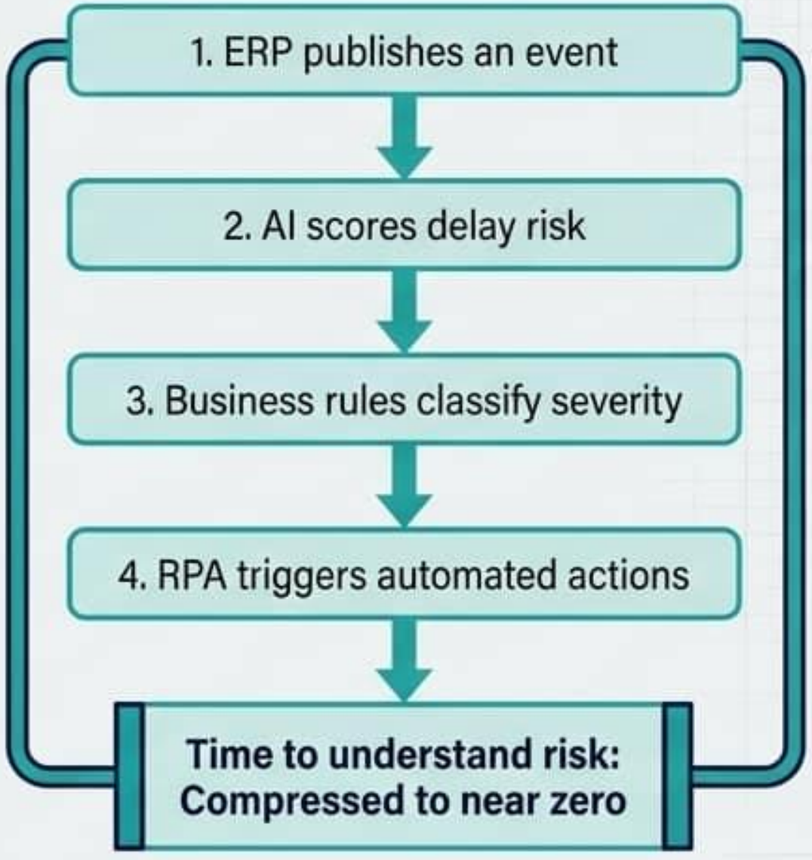
NotebookLM

AI creates measurable value only when it fundamentally redesigns operational workflows

BEFORE - REACTIVE FLOW

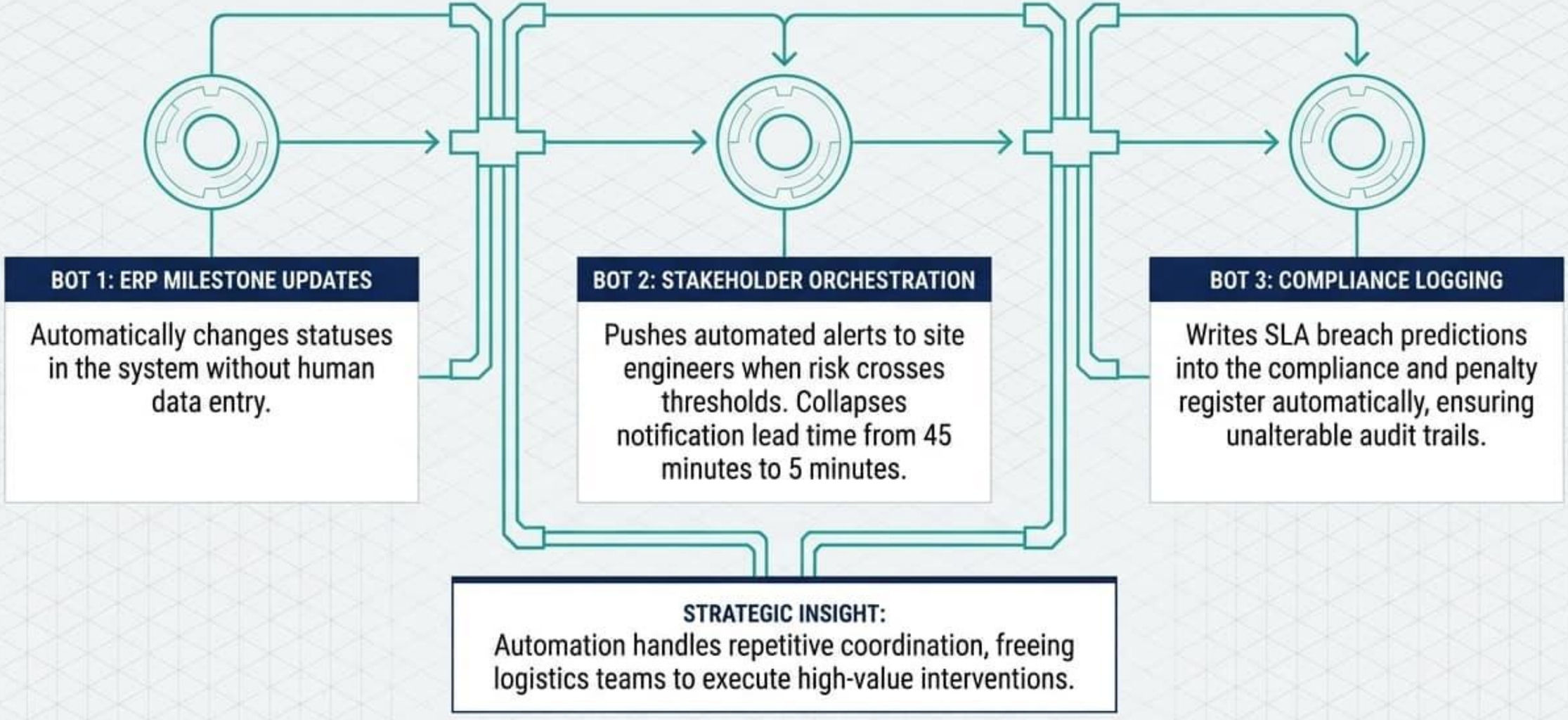


AFTER - EVENT-DRIVEN FLOW



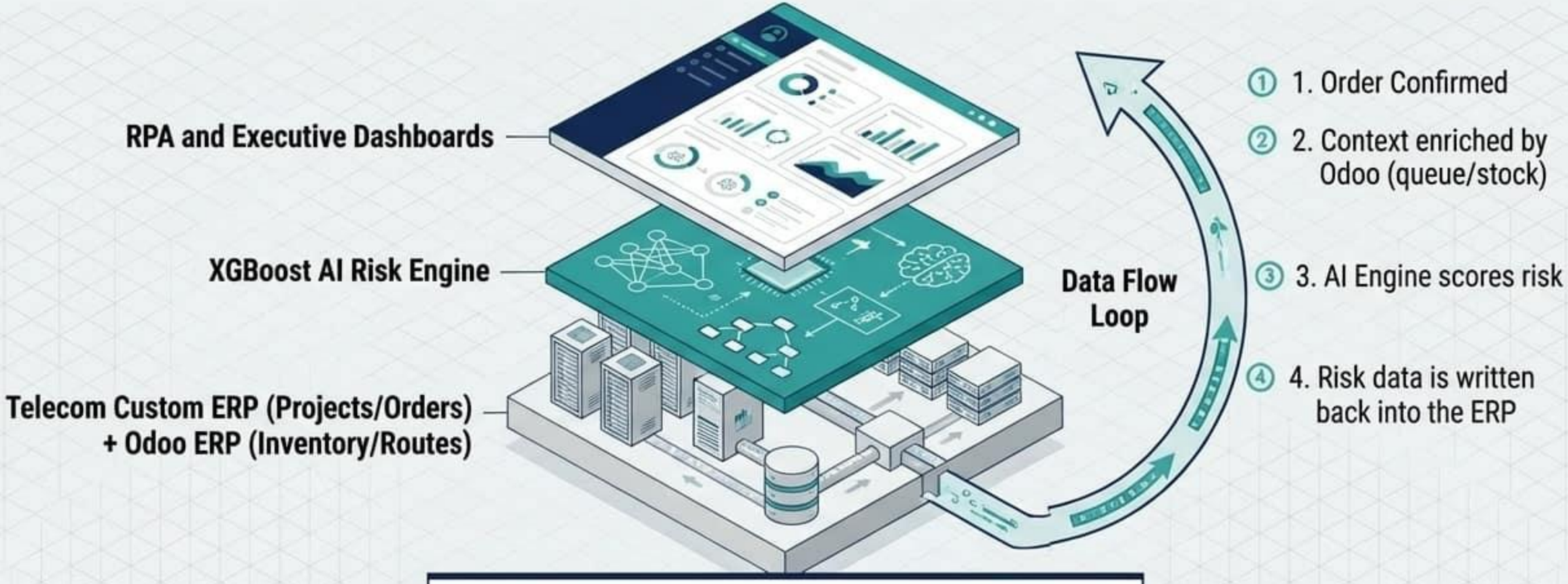
NotebookLM

Robotic Process Automation removes coordination friction to empower human exception management



NotebookLM

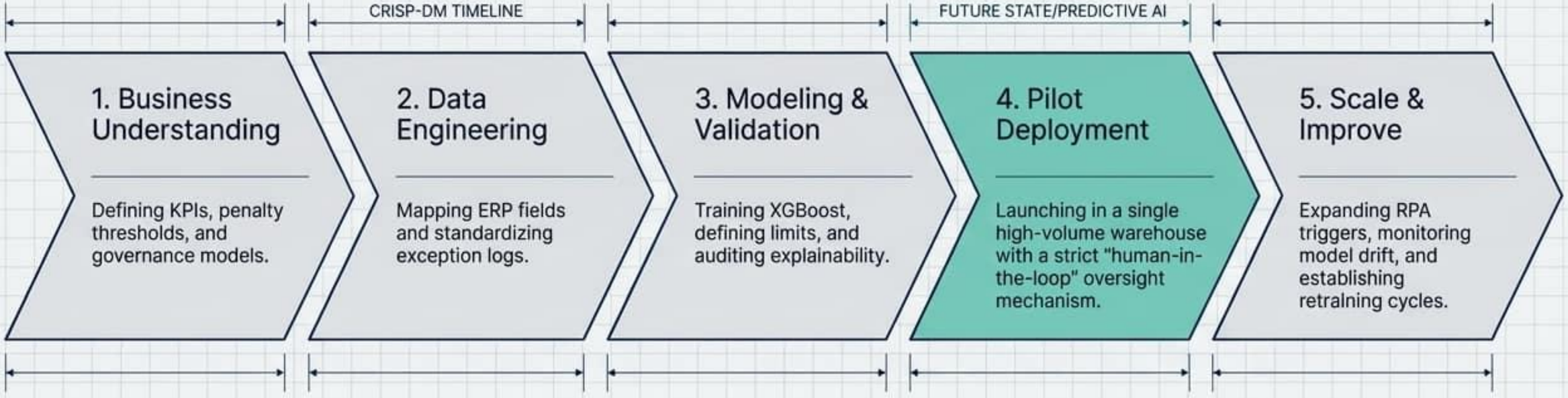
True capability requires embedding predictive intelligence directly into enterprise transactions



SYNTHESIS INSIGHT:
Prediction without integration is merely a dashboard; prediction wired directly into ERP logic is an enterprise capability.

NotebookLM

A Hybrid Agile rollout marries exploratory machine learning with strict enterprise governance



NotebookLM

AI-driven prioritization transforms a logistics cost-center into a highly optimized operation

20.5% → 12.3%

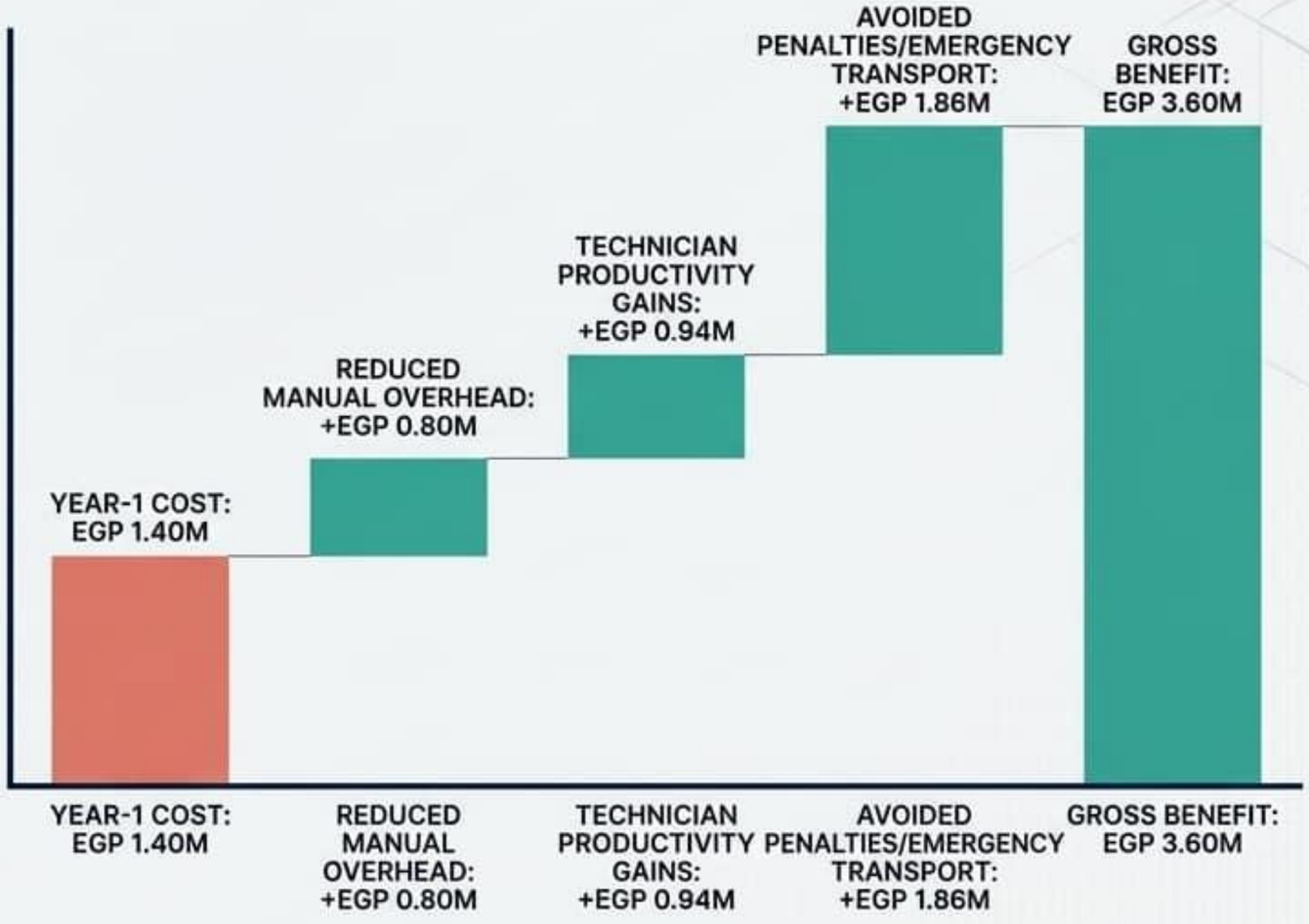
SLA BREACH RATE DROP

1,199

AVOIDED BREACHES ANNUALLY

157%

INDICATIVE YEAR-1 ROI WITH <6 MONTH PAYBACK



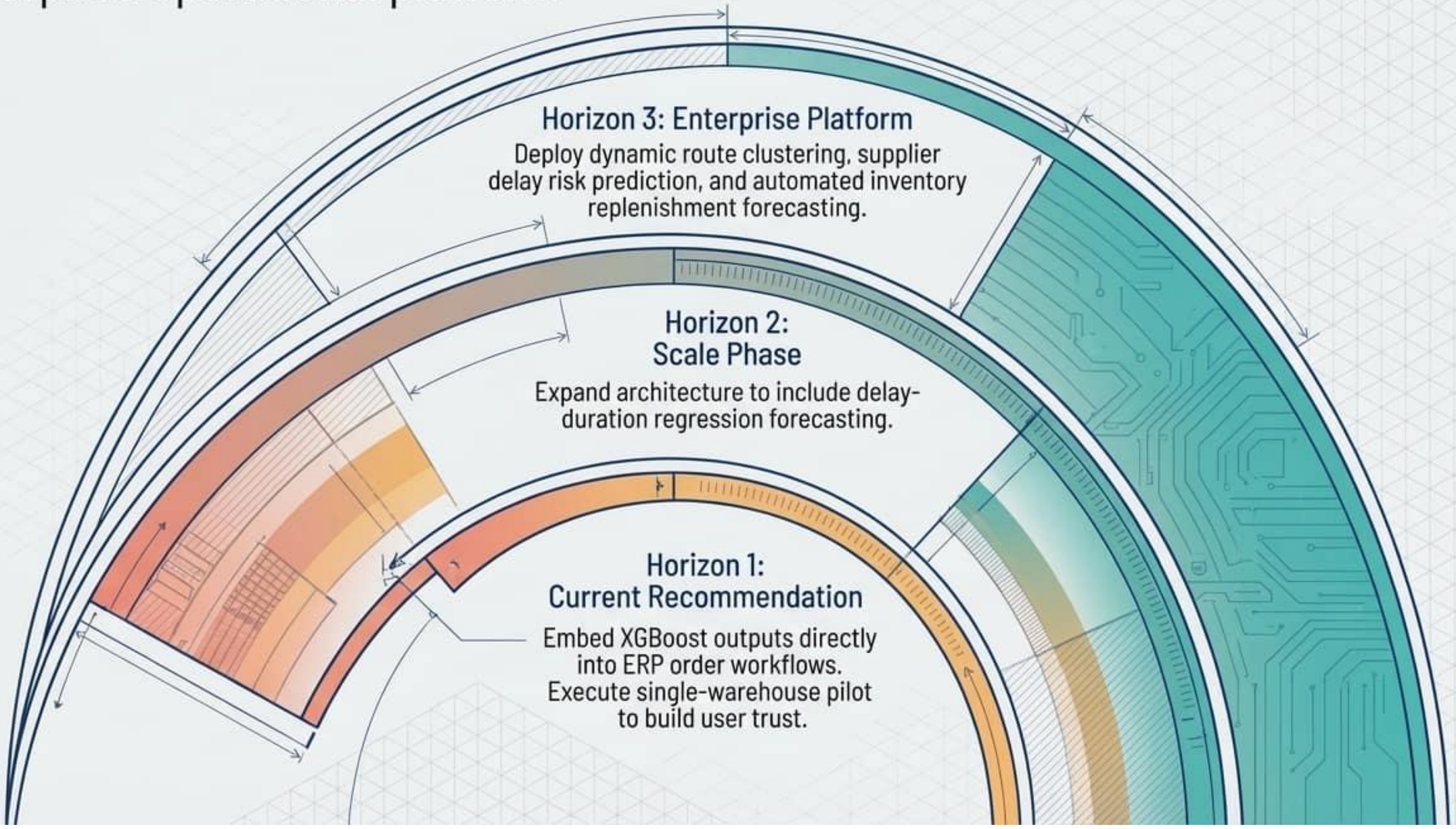
NotebookLM

Model performance requires uncompromising adherence to privacy, security, and fairness controls

PRIVACY (GDPR Alignment)	Strict data minimization regarding employee and customer identifiers in model training sets to ensure purpose limitation.
SECURITY (ISO/IEC 27001)	Implementation of Clauses 6, 8, and 9. Enforcing strict API access management, automated action logging, and rigorous supplier security.
ETHICS & ALGORITHMIC BIAS	Mandating 'human-in-the-loop' override capabilities to prevent historically biased prioritization metrics from becoming hardcoded automated law.

NotebookLM

Delay prediction forms the foundational node of an intelligent enterprise operational platform



NotebookLM

Synthesis is grounded in academic rigor, industry standards, and empirical data mining.

METHODOLOGY FOUNDATION

Applied CRISP-DM methodology, utilizing a robust simulated dataset comprising **14,600 delivery records** designed to strictly mirror realistic telecommunications operations.

CORE REFERENCES

Akbari & Do (2021): ML in logistics and supply chain trends.

Davenport & Brooks (2004): Enterprise systems, ERP, and supply chain value creation.

ISO/IEC 27001:2022: Information security management systems frameworks.

Odoo S.A. Architecture Documentation: Replenishment logic and warehouse pushing/pulling rules.

NotebookLM